MTH 310
Spring, 2026

Exam 1

Name:____________________
Sections 12.1–14.3

Instructions. (100 points) Sections 12.1–14.3. You may use a calculator. Show all work for credit.

(10pts) **1.** Let $A = (1, 3, -2)$, $B = (4, 1, 2)$, and $C = (2, 5, 0)$.

Find the scalar projection of $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

Solution:

$\overrightarrow{AB} = \langle 3, -2, 4\rangle, \quad \overrightarrow{AC} = \langle 1, 2, 2\rangle$

$\overrightarrow{AC} \cdot \overrightarrow{AB} = 3 - 4 + 8 = 7$

$|\overrightarrow{AB}| = \sqrt{9 + 4 + 16} = \sqrt{29}$

$$\text{comp}_{\overrightarrow{AB}}\overrightarrow{AC} = \frac{\overrightarrow{AC} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{7}{\sqrt{29}}$$

(12pts) **2.** Consider the two lines:

$$L_1: \quad x = 2 + t, \quad y = 1 - t, \quad z = 3t$$
$$L_2: \quad x = 4 + 2s, \quad y = -1 - 2s, \quad z = 6s$$

Are the lines **parallel**, **intersecting**, or **skew**? Justify your answer completely:

- If they are parallel, determine whether they are the **same line** or **two distinct lines**.
- If they intersect, find the point of intersection.

Solution:

Direction vectors: $\mathbf{d_1} = \langle 1, -1, 3\rangle, \quad \mathbf{d_2} = \langle 2, -2, 6\rangle = 2\mathbf{d_1}$.

Since $\mathbf{d_2}$ is a scalar multiple of $\mathbf{d_1}$, the lines are **parallel**.

Now check if they are the same line. L_1 passes through $(2, 1, 0)$ (set $t = 0$). L_2 passes through $(4, -1, 0)$ (set $s = 0$).

Is $(4, -1, 0)$ on L_1? Solve $2 + t = 4 \Rightarrow t = 2$. Then $y = 1 - 2 = -1$ ($\checkmark$) and $z = 6 \neq 0$ ($\times$).

Since $(4, -1, 0) \notin L_1$, the lines are **parallel but distinct**.

22 pts

(16[pts]) **3.** Consider the surface $4x^2 - y^2 + 4z^2 = 16$.

(a) *(8 pts)* Find the family of traces in the most important direction and at least one trace in each of the other two coordinate directions. Use these to give a complete description of the surface.

Solution:

Divide by 16: $\dfrac{x^2}{4} - \dfrac{y^2}{16} + \dfrac{z^2}{4} = 1$

Trace $y = k$ (most important — the family): $\frac{x^2}{4} + \frac{z^2}{4} = 1 + \frac{k^2}{16}$ — **circles** of radius $2\sqrt{1 + k^2/16}$ for all values of k. The circles grow as $|k|$ increases. The smallest circle is at $y = 0$ with radius 2.

Trace $z = 0$ (xy-plane): $\frac{x^2}{4} - \frac{y^2}{16} = 1$ — **hyperbola** opening in the x-direction, vertices at $(\pm 2, 0, 0)$.

Trace $x = 0$ (yz-plane): $\frac{z^2}{4} - \frac{y^2}{16} = 1$ — **hyperbola** opening in the z-direction, vertices at $(0, 0, \pm 2)$.

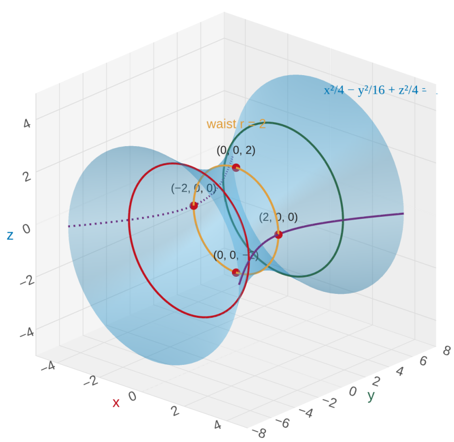
This is a **hyperboloid of one sheet** with axis along the y-axis. The waist (smallest circle) is at $y = 0$ with radius 2.

(b) *(8 pts)* Carefully sketch the surface, labeling the intercepts and the axis of symmetry.

Solution:

Intercepts: $(\pm 2, 0, 0)$ and $(0, 0, \pm 2)$. No y-intercepts.

Axis of symmetry: the y-axis. The surface looks like a cooling tower with circular cross-sections, narrowest at $y = 0$ and widening as $|y|$ increases.



16 pts

(16pts) **4.** Let $\mathbf{r}(t) = \langle 2\cos t,\ 2\sin t,\ 3t\rangle$.

(a) *(4 pts)* Show that the curve lies on a circular cylinder and give its equation.

Solution:

$x = 2\cos t$ and $y = 2\sin t$, so $x^2 + y^2 = 4\cos^2 t + 4\sin^2 t = 4$.

The curve lies on the cylinder $x^2 + y^2 = 4$ (radius 2, axis along z).

(b) *(7 pts)* Find the unit tangent vector $\mathbf{T}$ at $t = \pi/2$.

Solution:

$\mathbf{r}'(t) = \langle -2\sin t,\ 2\cos t,\ 3\rangle$

$|\mathbf{r}'(t)| = \sqrt{4\sin^2 t + 4\cos^2 t + 9} = \sqrt{4+9} = \sqrt{13}$

$\mathbf{r}'(\pi/2) = \langle -2, 0, 3\rangle$

$$\mathbf{T}(\pi/2) = \frac{\mathbf{r}'(\pi/2)}{|\mathbf{r}'(\pi/2)|} = \frac{1}{\sqrt{13}}\langle -2, 0, 3\rangle = \left\langle \frac{-2}{\sqrt{13}},\ 0,\ \frac{3}{\sqrt{13}} \right\rangle$$

(c) *(5 pts)* Find parametric equations for the tangent line to the curve at $t = \pi/2$.

Solution:

The point is $\mathbf{r}(\pi/2) = \langle 0, 2, 3\pi/2\rangle$ and the direction is $\mathbf{r}'(\pi/2) = \langle -2, 0, 3\rangle$.

$x = -2s, \quad y = 2, \quad z = \frac{3\pi}{2} + 3s$

16 pts

(12pts) **5.** Find a vector function $\mathbf{r}(t)$ that represents the curve of intersection of the cylinder $y^2+z^2 = 4$ and the surface $x = yz$.

Solution:

The cylinder $y^2 + z^2 = 4$ is parameterized by $y = 2\cos t,\ z = 2\sin t$.
Then $x = yz = (2\cos t)(2\sin t) = 4\cos t \sin t = 2\sin 2t$.

$$\mathbf{r}(t) = \langle 2\sin 2t,\ 2\cos t,\ 2\sin t\rangle, \quad 0 \le t \le 2\pi$$

12 pts

(12^{pts}) **6.** Consider the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x\,y^4}{x^4+y^4}$$

Does this limit **exist** or **not exist**? Provide careful evidence to support your conclusion. If it exists, prove it. If it does not exist, show why.

Solution:

The limit **exists and equals** 0.

Since $y^4 \le x^4 + y^4$, we have $\dfrac{y^4}{x^4+y^4} \le 1$.

Therefore:

$$\left|\frac{x\,y^4}{x^4+y^4}\right| = |x| \cdot \frac{y^4}{x^4+y^4} \le |x| \cdot 1 = |x|$$

As $(x,y) \to (0,0)$, $|x| \to 0$, so by the Squeeze Theorem:

$$\lim_{(x,y)\to(0,0)} \frac{x\,y^4}{x^4+y^4} = 0$$

12 pts

(12pts) **7.** For $g(r,s,t) = e^r \sin(st)$, find the mixed partial derivative g_{rst}.

Solution:

$g_r = e^r \sin(st)$

$g_{rs} = e^r \cdot t\cos(st)$

$g_{rst} = e^r\big[\cos(st) + t\big(-\sin(st)\big)\cdot s\big] = e^r[\cos(st) - st\sin(st)]$

(Product rule on $t\cos(st)$: derivative of t is 1, derivative of $\cos(st)$ w.r.t. t is $-s\sin(st)$.)

(10pts) **8.** The equation $xz + yz^2 = 6$ defines z implicitly as a function of x and y near the point $(1,1,2)$.

Find $\dfrac{\partial z}{\partial x}$ at $(1,1,2)$.

Solution:

Differentiate both sides with respect to x (treating y as constant, $z = z(x,y)$):

$z + x\frac{\partial z}{\partial x} + 2yz\frac{\partial z}{\partial x} = 0$

$\frac{\partial z}{\partial x}(x + 2yz) = -z \quad\Rightarrow\quad \frac{\partial z}{\partial x} = \frac{-z}{x+2yz}$

At $(1,1,2)$: $\frac{\partial z}{\partial x} = \frac{-2}{1+2(1)(2)} = \frac{-2}{5}$

22 pts